

Cluster 21525

To view the latest Map of Science clusters and data, visit the live Map at sciencemap.eto.tech.

Subjects

Research fields: Chemistry, Materials science, Engineering

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Inorganic chemistry, Chemical engineering, Physical chemistry

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: Methane Dehydroaromatization, methane conversion, active sites, non-oxidative methane, conversion of methane

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

1.10% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

382 recent articles in the cluster

382 articles in this cluster were published in the five year period ending in 2023.

Average article is **14.69** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **9.43th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 9.43% of other clusters.

Citation rating: **60.65th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 60.65% of all other articles published in the same year.

87.70% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 339 of 382 articles. 299 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	99	2.53
United States	58	3.82
South Korea	35	2.86
Canada	30	2.09
Netherlands	22	4.18
Russia	21	0.71
Japan	18	1.81
Uzbekistan	17	0.46
France	15	4.20
India	13	1.29

Top funding countries

Data available for 225 of 382 articles. 204 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	70	3.60
United States	38	4.70
South Korea	28	3.14
Netherlands	21	2.60
Canada	19	2.74
Japan	14	1.52
Germany	10	3.10
Saudi Arabia	9	3.85
Spain	9	7.23
Russia	8	0.86

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[Methane activation by ZSM-5-supported transition metal centers](#)

2020: Chemical Society reviews. 93 citations.

[Catalytic Routes for Direct Methane Conversion to Hydrocarbons and Hydrogen: Current State and Opportunities](#)

2022: ACS Catalysis. 35 citations.

[Coking mechanism of Mo/ZSM-5 catalyst in methane dehydroaromatization](#)

2021: Applied Catalysis A: General. 34 citations.

[A critical literature review of the advances in methane dehydroaromatization over multifunctional metal-promoted zeolite catalysts](#)

2020: Applied Catalysis A: General. 71 citations.

[Reactivity, Selectivity, and Stability of Zeolite-Based Catalysts for Methane Dehydroaromatization](#)

2020: Advances in Materials. 122 citations.

[Dual Active Sites on Molybdenum/ZSM-5 Catalyst for Methane Dehydroaromatization: Insights from Solid-State NMR Spectroscopy](#)

2021: Angewandte Chemie International Edition. 54 citations.

[How do the products in methane dehydroaromatization impact the distinct stages of the reaction?](#)

2022: Applied Catalysis B: Environmental. 24 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[Heterogeneous catalysts for the non-oxidative conversion of methane to aromatics and olefins](#)

2021: Elsevier eBooks. 5 citations.

[Reaction Fundamentals and Reactor Configuration for Methane Dehydroaromatization: Present State and Future Prospects](#)

2024: Energy & Fuels. 1 citations.

[Catalytic Routes for Direct Methane Conversion to Hydrocarbons and Hydrogen: Current State and Opportunities](#)

2022: ACS Catalysis. 35 citations.

[Application of co-feeds in the catalytic conversion of light alkanes to aromatics: A comprehensive review](#)

2023: ChemCatChem. 4 citations.

[State-of-the-Art and Achievements in the Catalytic Conversion of Natural Gas into Valuable Chemicals](#)

2022: Catalysis in Industry. 3 citations.

[A critical literature review of the advances in methane dehydroaromatization over multifunctional metal-promoted zeolite catalysts](#)

2020: Applied Catalysis A: General. 71 citations.

[Reactivity, Selectivity, and Stability of Zeolite-Based Catalysts for Methane Dehydroaromatization](#)

2020: Advances in Materials. 122 citations.

[Assessment of the current state of research and achievements in the field of catalytic processing of natural gas into valuable chemical products](#)

2021: Катализ в промышленности. 2 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Methane activation by ZSM-5-supported transition metal centers](#)

2020: Chemical Society reviews. 93 citations.

[Activation and conversion of alkanes in the confined space of zeolite-type materials](#)

2021: Chemical Society reviews. 115 citations.

[A critical literature review of the advances in methane dehydroaromatization over multifunctional metal-promoted zeolite catalysts](#)

2020: Applied Catalysis A: General. 71 citations.

[Reactivity, Selectivity, and Stability of Zeolite-Based Catalysts for Methane Dehydroaromatization](#)

2020: Advances in Materials. 122 citations.

[Dual Active Sites on Molybdenum/ZSM-5 Catalyst for Methane Dehydroaromatization: Insights from Solid-State NMR Spectroscopy](#)

2021: Angewandte Chemie International Edition. 54 citations.

[Unravelling the Enigma of Nonoxidative Conversion of Methane on Iron Single-Atom Catalysts](#)

2020: Angewandte Chemie International Edition. 62 citations.

[Programmable heating and quenching for efficient thermochemical synthesis](#)

2022: Nature. 104 citations.

[Novel Strategy for the Synthesis of Ultra-Stable Single-Site Mo-ZSM-5 Zeolite Nanocrystals](#)

2020: Angewandte Chemie International Edition. 80 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 369 out of 382 articles. 103 of these articles came from these sources.

Source	Articles (last five years)
Chemical Engineering Journal	20
Fuel	13
Catalysis Today	11
Applied Catalysis A: General	11
ACS Catalysis	10
Angewandte Chemie International Edition	8
Catalysts	8
Applied Catalysis B: Environmental	8
Journal of Catalysis	7
DO DESENVOLVIMENTO MUNDIAL COMO RESULTADO DE REALIZAÇÕES EM CIÊNCIA E INVESTIGAÇÃO CIENTÍFICA - Volume 2	7

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 380 of 382 articles. 59 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Hua Song	24	4.79
Do Heui Kim	19	6.79
Zhaofei Li	19	5.06
Hao Xu	16	5.03
Yong Hyun Lim	15	6.62
Л. Л. Коробицына	15	0.56
А. А. Степанов	14	0.57
Hae Won Ryu	14	7.45
А. В. Восмери́ков	14	0.56
Shijun Meng	13	5.33

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 352 of 382 articles. 122 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
University of Calgary - Canada	25	2.34
Seoul National University - South Korea	19	3.29
Samarkand State University named after Sharof Rashidov - Uzbekistan	16	0.49
Institute of Petroleum Chemistry - Russia	14	0.62
Chinese Academy of Sciences - China	13	3.89
Самаркандский государственный университет	10	0.96
Indian Institute of Technology Delhi - India	9	1.59
West Virginia University - United States	9	3.9
Sogang University - South Korea	9	4.31
Eindhoven University of Technology - Netherlands	8	5.88

Organization types

This table covers the 320 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 382 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
3.45%	38.44%	12.04%	2.89%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 36 out of 382 articles. 22 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Total (Belgium)	5	5.86
Sinopec (China)	5	1
Leidos (United States)	4	6.96
Shell (Netherlands)	4	4.41
Shell (India)	2	0.73
Catalyse - France	2	5.5
Beijing Chemical Industry Research Institute (China)	1	3
Arkema (France)	1	2.75
Cummins (United States)	1	18.6
Total (Qatar)	1	0.5

Top funders

Data available for 250 out of 382 recent articles. 149 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	55	3.36
National Research Foundation of Korea - South Korea	25	3.31
United States Department of Energy - United States	23	4.62
Ministry of Science and ICT - South Korea	19	3.09
National Science Foundation - United States	18	5.60
Natural Sciences and Engineering Research Council - Canada	18	2.85
Ministry of Science, ICT & Future Planning, Republic of Korea	16	2.76
Ministry of Education - China	15	5.25
Kara Technologies Inc.	13	3.52
Dutch Research Council - Netherlands	12	2.62

Funder types

This table covers the 212 articles in this cluster from the last five years with data about their funding organization types (out of 382 articles total).

Commercial	Education	Nonprofit	Government
4.32%	9.19%	7.03%	79.46%

Patents and industry

246 patents cite articles in this cluster (**91.98th** percentile among all clusters)

15.52% of articles in the cluster are cited by patents

3.19% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
HYDROCARBON CONVERSION	9
SYSTEMS, METHODS, AND DEVICES FOR METHANE CONVERSION VIA GAS RECYCLING	33
CATALYST STRUCTURE AND METHOD OF UPGRADING HYDROCARBONS IN THE PRESENCE OF THE CATALYST STRUCTURE	32
ORGANIC SOLID BIOMASS CONVERSION FOR LIQUID FUELS/CHEMICALS PRODUCTION IN THE PRESENCE OF METHANE CONTAINING GAS ENVIRONMENT AND CATALYST STRUCTURE	28
METHOD OF LIGHT OIL DESULFURIZATION IN THE PRESENCE OF METHANE CONTAINING GAS ENVIRONMENT AND CATALYST STRUCTURE	28
DIRECT NON-OXIDATIVE METHANE CONVERSION IN A CATALYTIC WALL REACTOR	19
SYSTEMS, METHODS, AND DEVICES FOR DIRECT CONVERSION OF METHANE	19
SUPPORTED IRON CATALYSTS, METHODS OF MAKING, METHODS OF HYDROCARBON DECOMPOSITION	18
METHODS AND SYSTEMS FOR PRODUCING LIQUID HYDROCARBONS	14

Top patent assignees

Patent assignee	Same-cluster citations
EXXONMOBIL (UNITED STATES)	49
UNIVERSITY OF MARYLAND GLOBAL CAMPUS	46
SATURN LICENSING, LLC	36
SHIJUN MENG	32
加睿技术有限责任公司	32
HUA SONG	32
KARA TECHNOLOGIES INC	32
KARA ENERGY SERVICES INC	32
PENG HE	32

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
40112	97	photocatalytic methane conversion, methane oxidation, oxidation of methane, Selective Photocatalytic Oxidation, methane direct conversion
18830	68	catalysts, Lewis acid sites, zeolite, selectivity, BTX
2539	22	MTO reaction, zeolite catalysts, methanol conversion, acid sites, MTO process
7208	22	Oxidative Coupling, Coupling of Methane, OCM reaction, OCM catalysts, oxygen species
5423	15	DBD plasma reactor, plasma reforming process, catalyst, plasma catalysis, dry reforming
6557	10	light olefins, selectivity, carbon dioxide hydrogenation, Fe-based catalysts, carbon dioxide
19019	9	Hydrogen Production, methane pyrolysis, Energy Publications LLC, Catalytic methane decomposition, steam methane reforming
458	8	methane dry reforming, reforming of methane, Ni-based DRM catalysts, DRM reaction, carbon deposition
7505	8	catalytic pyrolysis, biomass pyrolysis process, aromatic hydrocarbons, lignocellulosic biomass, pyrolysis of biomass
10939	7	Transition metal carbides, molybdenum carbide, metal carbide catalysts, catalytic activity, supported metal catalysts

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
18830	53	catalysts, Lewis acid sites, zeolite, selectivity, BTX
7208	43	Oxidative Coupling, Coupling of Methane, OCM reaction, OCM catalysts, oxygen species
40112	40	photocatalytic methane conversion, methane oxidation, oxidation of methane, Selective Photocatalytic Oxidation, methane direct conversion
2539	39	MTO reaction, zeolite catalysts, methanol conversion, acid sites, MTO process
3414	34	Hierarchical zeolites, zeolite catalysts, catalytic performance, Mesoporous Zeolites, catalytic activity
4509	20	density functional theory, hydrogen evolution reaction, functional theory calculations, electronic structure, photocatalytic OER catalysts
19019	18	Hydrogen Production, methane pyrolysis, Energy Publications LLC, Catalytic methane decomposition, steam methane reforming
458	14	methane dry reforming, reforming of methane, Ni-based DRM catalysts, DRM reaction, carbon deposition
10939	14	Transition metal carbides, molybdenum carbide, metal carbide catalysts, catalytic activity, supported metal catalysts
5715	13	microwave heating, Microwave Irradiation Technique, Microwave Assisted Synthesis, Microwave Nonthermal Effect, traveling-wave microwave reactor

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
Project-Platypus/Platypus	1	571
irenamal/EC-KitY	1	0
jenetics/jenetics	1	850
geatpy-dev/geatpy	1	2010
EC-KitY/EC-KitY	1	92
trevorstephens/gplearn	1	1605